

Rheumatoid arthritis is caused by a Proteus urinary tract infection.

Genetic, molecular and biological studies indicate that rheumatoid arthritis (RA), a severe arthritic disorder affecting approximately 1% of the population in developed countries, is caused by an upper urinary tract infection by the microbe, *Proteus mirabilis*. Elevated levels of specific antibodies against *Proteus* bacteria have been reported from 16 different countries. The pathogenetic mechanism involves six stages triggered by cross-reactive autoantibodies evoked by *Proteus* infection. The causative amino acid sequences of *Proteus* namely, ESRRAL and IRRET, contain arginine doublets which can be acted upon by peptidyl arginine deiminase thereby explaining the early appearance of anti-citrullinated protein antibodies in patients with RA. Consequently, RA patients should be treated early with anti-*Proteus* antibiotics as well as biological agents to avoid irreversible joint damages. © 2013 APMIS. Published by John Wiley & Sons Ltd.